Scalar Hall Exclusion in Dimensions 11 through 15: A Partial Lean Formalization

Ryosuke Shiroshita
Ryosuke.Shiroshita@exactory.ai

Abstract

We formalize a finite scalar layer of an existing computer-assisted argument for Dittert's conjecture. For dimensions 11 through 15, the artifact checks direct rational formulas for every ordered proper cut and one normalized marginal cut, then derives a strict polynomial inequality for every real parameter $u \leq 1$. Its corollary excludes simultaneous permanent and deficit lower bounds even under a relaxed scalar budget. Lean verifies 340 normalized cases and proves that they cover the entire specified nonfull scalar domain. Independent execution and statement review confirm that the positivity, exclusion, and finite-certificate declarations depend only on `propext`, `Classical.choice`, and `Quot.sound`. A separate appendix verifies two literal near-zero polynomial certificates in dimensions 8 and 9 on specified closed intervals; it does not establish full scalar coverage in either dimension. The contribution is an attributed partial formalization and reproducible verification artifact. The matrix reduction, its analytic and structural inputs, and the full Dittert theorem are outside the formalization.

1 Source argument and contribution

For a nonnegative real $n \times n$ matrix A of total entry sum n, let r_i and c_j be its row and column sums. Dittert's functional is

$$\Phi(A) = \prod_{i=1}^{n} r_i + \prod_{j=1}^{n} c_j - \operatorname{per}(A).$$

The conjecture asserts $\Phi(A) \leq 2 - n!/n^n$, with equality precisely at the matrix whose entries are $1/n$. Cheon and Wanless establish structural restrictions on potential maximizers, including exclusions associated with partial decomposability and rectangular zero sets [2]. Pang's scaling argument provides related large-dimension context [7].

This note concerns a scalar calculation in Nascimento's public working proof, at a fixed source revision.[1] That repository already presents an assembled working-proof chain and states the review boundaries of its components. Its Hall argument associates a possible boundary maximizer with cut sizes and a scaling parameter, bounds its permanent from below, and bounds its deficit from below. A strict scalar inequality makes those bounds incompatible with the available budget. The finite arithmetic and this final scalar implication are the objects formalized here.

The source mathematics is not a new discovery of this artifact. In particular, the note does not claim a new mathematical theorem or the first formalization of Dittert's conjecture. Li, Xiong, and Yang already report a dimension-four proof based on an exact sum-of-squares certificate and a Lean formalization [4]. Our scope is a different, explicitly bounded scalar layer of the public Hall

[1] Pedro Paulo Marques do Nascimento, `pedromnasc/dittert-conjecture-proof`,
Source commit: `894066bbaa715138c98bf3cb7c6fdb4f39a37701`.

argument. Neither the name of a scalar variable nor a successful arithmetic check establishes its relationship to an arbitrary matrix.

The formalization uses Lean 4 [3] and Mathlib [5]. The implementation combines direct dimension-dependent formulas, exhaustive coverage, rational-to-real transport, a completed-square argument, Bernoulli comparison, and the relaxed-budget corollary. The accompanying evidence records both the exact theorem and what remains outside it.

2 Exact scalar statement

All factorials and powers below have natural indices. The constants are first defined in $\mathbb{Q}$ and then mapped into $\mathbb{R}$ by the canonical embedding. For the positive indices used below, define

$$\gamma_k = \frac{k!}{k^k}. \tag{1}$$

Fix $n, a, b \in \mathbb{N}$ satisfying

$$11 \le n \le 15, \qquad 0 < a + b < n. \tag{2}$$

Put $p = n - a - b$. The scalar floor and curvature are

$$M_{n,a,b} = \begin{cases} \dfrac{\gamma_{n-a}\gamma_{n-b}}{\gamma_{n-a-b}}, & a > 0,\ b > 0, \\ (n-2)!\left(\dfrac{n-2}{(n-1)^2}\right)^{n-2}, & a = 0 \text{ or } b = 0, \end{cases} \tag{3}$$

and

$$C_{n,a,b} = \begin{cases} \dfrac{7n(n-a-b)^2}{16\{a(n-a)+b(n-b)\}}, & a > 0,\ b > 0, \\ \dfrac{7n}{16(n-1)}(1+(1-\gamma_n)^2), & a = 0 \text{ or } b = 0. \end{cases} \tag{4}$$

The domain ensures that p, $n - a$, and $n - b$ are positive. Every denominator displayed in the proper branch is therefore positive; the marginal branch has $n - 1 > 0$. Natural subtraction in the Lean definitions agrees with ordinary subtraction at all these indices. Although the definitions are total Lean functions, no assertion here uses out-of-domain values to avoid a denominator obligation.

Theorem 1 (Scalar positivity)**.** *For every $n, a, b \in \mathbb{N}$ satisfying (2) and every $u \in \mathbb{R}$ with $u \le 1$,*

$$C_{n,a,b}u^2 + M_{n,a,b}(1-u)^n - \gamma_n > 0. \tag{5}$$

Corollary 2 (Relaxed-budget exclusion)**.** *Under the same natural-size restrictions, there are no real u, P, δ with $u \le 1$ satisfying all three inequalities*

$$P \ge M_{n,a,b}(1-u)^n, \qquad \delta \ge C_{n,a,b}u^2, \qquad P + \delta \le \gamma_n. \tag{6}$$

Proof. The first two inequalities imply $P+\delta-\gamma_n \ge M_{n,a,b}(1-u)^n + C_{n,a,b}u^2 - \gamma_n > 0$ by Theorem 1, contradicting the last inequality. □

There is no lower bound on u in either statement. The polynomial result includes both endpoints of $[0, 1]$ and every negative u. This larger scalar domain does not assert that the matrix construction has a corresponding interpretation there. Likewise, P and δ are arbitrary real variables in Corollary 2; no definition identifying them with matrix quantities occurs in the theorem.

3 Finite criterion and real-variable proof

The finite calculation proves a strict discriminant condition. Writing $M = M_{n,a,b}$ and $C = C_{n,a,b}$, the exact predicate is

$$M > 0, \qquad C > 0, \qquad D_{n,a,b} := 4C(M - \gamma_n) - (nM)^2 > 0. \tag{7}$$

This is the conjunction named `HallScalar.Good` in the source. It is checked directly from (1)–(4), without a table of externally computed successful margins as a premise.

Lemma 3 (Exhaustive finite criterion)**.** *Every triple satisfying* (2) *satisfies* (7)*.*

Verification and coverage. Call (a, b) proper if $a, b > 0$ and $a+b < n$. At each of the five dimensions, `Finite.lean` proves the implication from the reduced predicate to `Good` for every (a, b) in `Fin 15` $\times$ `Fin 15`. The reduced predicate consists of all ordered proper pairs and the pair $(n-1, 0)$. The five declarations `finite_n11` through `finite_n15` use `decide +kernel`. Thus Lean reduces the finite decidable proposition using its ordinary rational arithmetic definitions.

For an arbitrary proper pair in (2), both coordinates are less than 15, so the pair lies in the checked universe. The declaration `reduced_good` proves this embedding and selects the applicable dimension. If a coordinate is zero, both piecewise formulas are independent of the other coordinate. Consequently their values, and hence the predicate `Good`, equal those at $(n-1, 0)$. The declaration `all_good` makes this split and covers every natural pair in the domain. No transpose quotient is taken for proper pairs, and neither marginal orientation is omitted. □

The finite statement is an exact computational lemma. Its proof is the checked reduction and coverage argument just described, rather than a printed decimal estimate for the weakest case. The complete rational formulas and quantified checks are included in the portable source.

Lemma 4 (Quadratic comparison)**.** *Let* $C, M, g, u \in \mathbb{R}$ *and* $n \in \mathbb{N}$*. If* $C > 0$ *and* $4C(M-g) - (nM)^2 > 0$*, then*

$$Cu^2 - nMu + M - g > 0.$$

Proof. The identity

$$4C(Cu^2 - nMu + M - g) = (2Cu - nM)^2 + 4C(M-g) - (nM)^2$$

has a nonnegative square and a strictly positive remainder on its right-hand side. Division by the positive factor $4C$ proves the claim. In Lean, `ring` checks the identity and ordered-field reasoning supplies the sign conclusion. □

Proof of Theorem 1. Lemma 3, transported from $\mathbb{Q}$ to $\mathbb{R}$, supplies $M > 0$, $C > 0$, and $D_{n,a,b} > 0$. Lemma 4 therefore gives $Cu^2 - nMu + M - \gamma_n > 0$ for every real u. For $u \le 1$, Bernoulli's inequality gives

$$(1-u)^n \ge 1 - nu.$$

Its usual hypothesis $-u \ge -1$ follows directly from $u \le 1$. Multiplication by $M > 0$ and addition of $Cu^2 - \gamma_n$ yield

$$Cu^2 + M(1-u)^n - \gamma_n \ge Cu^2 - nMu + M - \gamma_n > 0.$$

The formal proof invokes Mathlib's `one_add_mul_le_pow`; the installed version permits the weaker hypothesis $-2 \le -u$, which also follows from the stated domain. No interval sampling, root selection, or small-parameter assumption is used. □

4 Coverage and formal artifact

For a fixed n, the number of ordered proper pairs is

$$\sum_{s=2}^{n-1}(s-1) = \frac{(n-1)(n-2)}{2}.$$

The nonfull marginal pairs are $(a, 0)$ and $(0, b)$ with $1 \le a, b \le n-1$, giving $2(n-1)$ pairs. They share one scalar formula. Table 1 distinguishes these counts from the reduced family actually checked.

n	Proper	Marginal	Normalized marginal	Reduced	Uncompressed
11	45	20	1	46	65
12	55	22	1	56	77
13	66	24	1	67	90
14	78	26	1	79	104
15	91	28	1	92	119
Total	335	120	5	340	455

Table 1: Ordered nonfull scalar cases. "Reduced" adds one normalized marginal to every proper pair. "Uncompressed" retains every marginal size and orientation.

The declarations `case_counts` and `total_case_count` prove the five reduced counts and their sum 340 in Lean. The combined declaration `finite_certificate` includes both universal coverage and this cardinality result. The uncompressed total 455 is the combinatorial calculation above, checked in the independent source review; it is not presented as a named Lean cardinality theorem. Universal scalar coverage follows from `all_good`, regardless of how the explanatory table is computed.

The source modules separate the obligations. `Definitions.lean` contains the rational formulas and predicates. `Finite.lean` proves the finite criterion and coverage. `Main.lean` proves `HallScalar.scalar_positive` and `HallScalar.scalar_exclusion`. Finally, `Audit.lean` exposes `hall_scalar_certificate`, the same fully quantified exclusion corollary, through the test import chain.

The fixtures in `Tests.lean` check the normalization $\gamma_3 = 2/9$, proper and excluded cut boundaries, the normalized marginal, and exclusion of $(0, 0)$ from the reduced predicate. They also instantiate positivity at $u = 0$, $u = 1$, and $u = -2$, including a non-normalized marginal in the opposite orientation. These fixtures supplement the universal proof; they do not replace its coverage argument.

5 Verification and reproducibility

The portable package under `evidence/lean/` pins Lean `v4.33.1` and Mathlib revision

`0df444a360eaa60ab8c11dca51a86af692955474`.

Its manifest contains repository URLs and dependency revisions, without an absolute local dependency path. With the required toolchain and dependencies available, the package is built from that directory by

```
export LEAN_NUM_THREADS=1
nice -n 15 lake build
nice -n 15 lake env lean -j1 -M2048 ../FinalScalarAxiomAudit.lean
```

The configuration supplies one Lean thread and a 2048 MB Lean memory ceiling. A fresh environment may require the dependency cache download; the recorded verification reused existing dependencies. The harness also uses a lightweight axiom printer without those explicit command-line limits; the separate direct audit above applies them.

Independent source and execution review reran the full scalar package and the finite-only package, and directly audited the positivity theorem as well as its corollary. Both harness records report `pass`. These reviews were performed by separate assistant agents, not by external human peer reviewers. The four audited declarations were

```
HallScalar.scalar_positive
HallScalar.scalar_exclusion
HallScalar.finite_certificate
hall_scalar_certificate
```

Each printed exactly the transitive axiom list

```
[propext, Classical.choice, Quot.sound]
```

The accompanying `FinalScalarAxiomAudit.lean` checks the theorem types and prints these closures and the imported arithmetic definitions. The completed dependency graph contains no `sorryAx`, custom axiom, or native-evaluation axiom. Rational inequalities are converted to real inequalities using the standard cast interface; no arithmetic operation is locally redefined.

The finite-only package uses the same definitions and finite proof as the full package. It provides a separately executed audit of a different target, not an independent arithmetic implementation. Independence here refers to the reviewer and fresh executions, not to two distinct proof algorithms. The evidence includes source hashes checked before and after review. The review did not rebuild all Mathlib dependencies from source or use an alternate proof-assistant kernel.

Table 2 records two observed coordinator runs from `resource-measurements.json`. Both used macOS arm64, one Lean thread, niceness 15, and the stated memory ceiling. The rows have different incremental build states and shared machine load, so they are observations rather than a performance comparison. They do not measure total development effort, a clean installation, or total cost.

Observed build and axiom check	Wall seconds	Maximum RSS, bytes
Standalone finite package, dependencies cached	36.05	1 713 913 856
Completed real theorem, finite proofs cached	20.74	1 534 099 456

Table 2: Two recorded incremental runs under shared local load. These are not clean-build benchmarks.

6 Related work and remaining obligations

Exact certificate replay is an established formal-verification approach. Siddique and Mian describe independent rational replay and a sound Lean checker for geometric non-embeddability certificates [8]. That work provides a methodological comparison for making the certificate interpretation and trusted base explicit, rather than a runtime baseline for this finite family. The prior Dittert formalization of Li, Xiong, and Yang uses an SOS certificate [4]; this note verifies a Hall scalar comparison and does not reimplement their dimension-four proof.

Bernstein polynomials and de Casteljau computation have prior formal treatments [1, 6]. Appendix A separately verifies two literal expansions in an unnormalized Bernstein basis. It uses direct polynomial identities and positivity proofs, not a generic Bernstein checker, subdivision theorem, or de Casteljau implementation. These additional certificates are not used in Theorem 1.

The boundary of the formalization is substantive. It does not construct a Hall minorant from a matrix, prove the relevant permanent floors, derive row or column deficit estimates, or justify stationarity and support reductions. Marginal normalization proves equality of the defined scalar constants; the source argument selecting a smallest row of a matrix remains outside Lean. The full cut $(a, b) = (0, 0)$ is excluded by (2), and its separate analytic exclusion is not formalized here. Nor does the artifact prove the matrix inequality's equality characterization.

Complete coverage of the nonfull scalar domain is proved here in dimensions 11 through 15. The appendix adds only two near-zero polynomial results in dimensions 8 and 9. Full scalar coverage in dimensions 7 through 10, dimensions 5 and 6, and any extension above 15 remain outside this artifact. Lower-dimensional source arguments may involve different interfaces, including work attributed to Lu in the public proof chain; they cannot be inferred from the scalar theorem here.[2] Formalizing additional matrix interfaces or a different certificate family would require a separately stated result, checked coverage, and a corresponding extension of the evidence record. The present note claims the completed finite scalar formalization and the separately bounded polynomial certificates, with their documented verification boundaries only.

A Separate near-zero certificates in dimensions 8 and 9

This appendix concerns only the near-zero polynomial portion of the source marginal arguments. For $n \in \{8, 9\}$, use $\gamma_n = n!/n^n$ and define the following real constants and polynomials:

$$\begin{aligned} \kappa_n &= (n-2)!\left(\frac{n-2}{(n-1)^2}\right)^{n-2}, & L_n &= \kappa_n\left(1+\frac{1}{2n^2}\right), \\ E_n(u) &= 1-(2-\gamma_n)u, & H_n(u) &= (1-\gamma_n)u, \\ T_n(u) &= E_n(u) - \frac{H_n(u)}{n-1}, & F_n(u) &= (1-u)\left(1+\frac{u}{n-1}\right)^{n-1}. \end{aligned}$$

The literal polynomial checked in `NearZero.p` is

$$\begin{aligned} p_n(u) = {} & \big(2-\gamma_n + L_n(1-u)^n - F_n(u)\big)E_n(u)^n \\ & - \big(E_n(u)+H_n(u)\big)T_n(u)^{n-1}. \end{aligned} \tag{8}$$

There is no division by $E_n(u)$ in this definition and thus no nonvanishing premise for E_n is omitted.

Theorem 5 (Literal near-zero positivity)**.** *For every real u, $0 \le u \le 1/400$ implies $p_8(u) > 0$, and $0 \le u \le 1/500$ implies $p_9(u) > 0$.*

Proof. For $(n, d) = (8, 400)$ and $(9, 500)$, the supplied rational coefficients $c_{n,k} > 0$ give the exact identities, for every real x,

$$p_n(x/d) = \sum_{k=0}^{2n} c_{n,k} x^k (1-x)^{2n-k}. \tag{9}$$

[2] Hongyuan Lu's `lueluelue2006/dittert-n7-extension`,
Source commit: `56397b70299278b338a286b90361180833dbf983`.

There are respectively 17 and 19 coefficients, with homogeneous basis degrees 16 and 18. The binomial factors of the normalized Bernstein basis are absorbed into $c_{n,k}$. The complete rational lists are the literal constants in `Certificate8.lean` and `Certificate9.lean`, under `evidence/near-zero/`. For example, their initial coefficients are

$$c_{8,0} = \frac{77405634405}{1814205196009472}, \qquad c_{9,0} = \frac{33970014498175}{2629465015396073472}.$$

The declarations `NearZero.identity8` and `NearZero.identity9` prove (9) by unfolding (8), normalizing factorials and rational arithmetic with `norm_num`, and checking the full identity with `ring`.

If $0 \le x < 1$, all summands are nonnegative and the $k = 0$ summand is strictly positive. At $x = 1$ only the last summand survives, with value $c_{n,2n} > 0$. The declarations `NearZero.positive8` and `NearZero.positive9` check these signs exactly, including both endpoints. Finally, $0 \le u \le 1/d$ implies $0 \le du \le 1$, and $(du)/d = u$. This substitution proves the conjunction named `near_zero_certificate` in `Main.lean`. Neither the coefficient signs nor the identities are hypotheses of that theorem. □

The Python generator proposes constants and source text; it is outside the trusted proof. A separate assistant-agent review reran and audited the final conjunction and the four identity and positivity declarations, each with exactly `propext`, `Classical.choice`, and `Quot.sound`. The portable package uses the same pinned Lean and Mathlib versions as the scalar package. Its `replay.sh` rebuilds the proof modules and runs the five-declaration axiom audit; its README gives the standalone command and dependency requirements.

These certificates do not prove the remaining support intervals, their monotone tails, support compression, matrix reductions, or an $n = 7$ case. In particular, they are neither full dimension-8 or dimension-9 scalar coverage nor full Dittert proofs. Their contribution is the reproducible formal verification of these two existing, explicitly bounded polynomial inequalities.

References

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